

Evaluating frontier agents on **economically valuable** tasks.

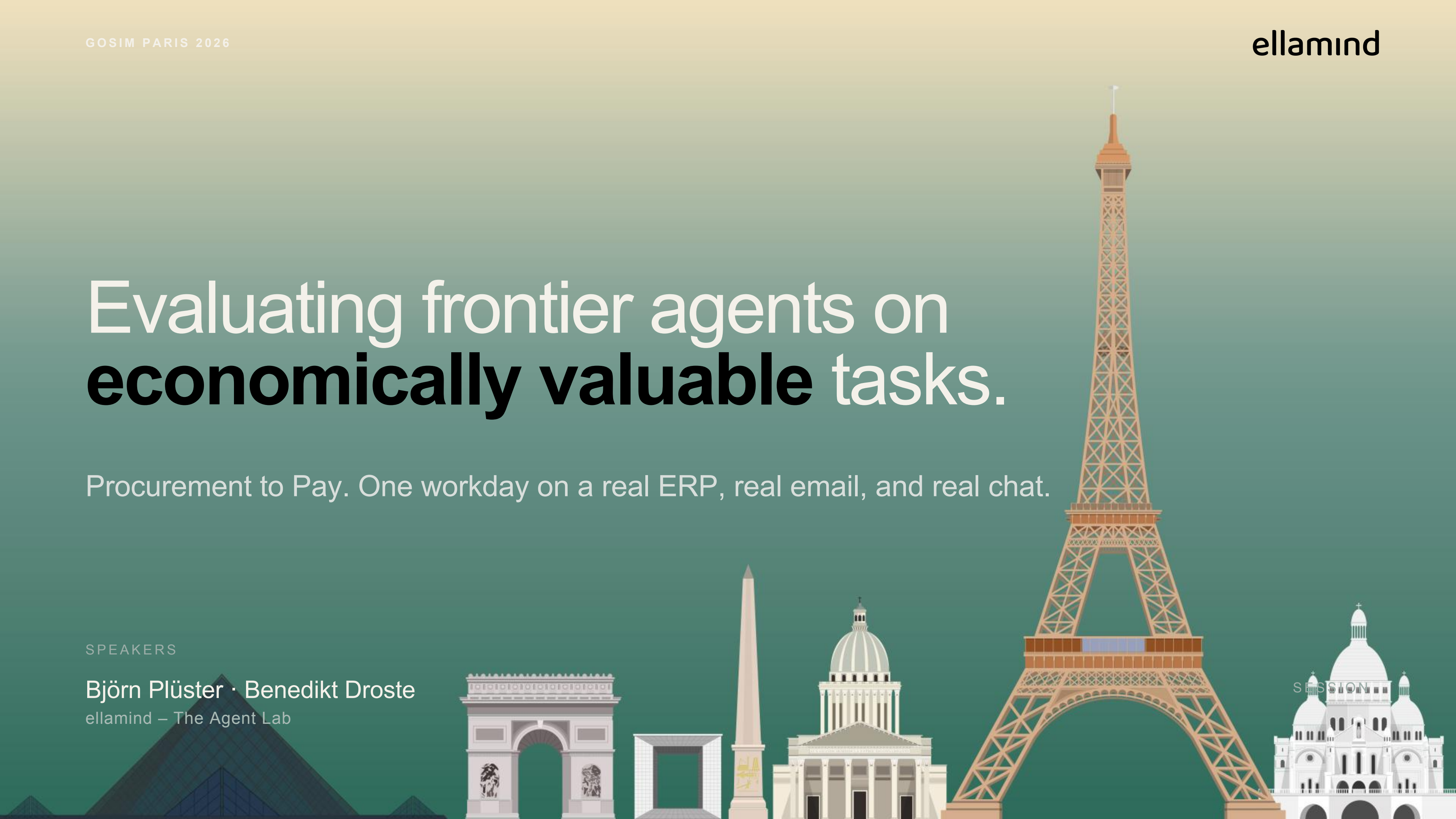
Procurement to Pay. One workday on a real ERP, real email, and real chat.

SPEAKERS

Björn Plüster · Benedikt Droste

ellamind – The Agent Lab

SESSION



01 STARTING POINT

Capability is high. Adoption keeps stalling.

GPT-5.5 • LAUNCH BENCHMARKS

Nine launch benchmarks: coding, search, math, tool use, computer use. Published 2026-04-23.

Terminal-Bench 2.0	82.7
Expert-SWE • internal	73.1
GDPval	84.9
OSWorld-Verified	78.7
Toolathlon	55.6
BrowseComp	84.4
FrontierMath • T1–3	51.7
FrontierMath • T4	35.4
CyberGym	81.8

OPENAI.COM/INDEX/INTRODUCING-GPT-5-5

ADOPTION • THREE READS, SAME SHAPE

Two-thirds piloting, a sliver in production. The gap is not model quality.

14%

of pilots reach production

78 % of enterprises now have pilots; only 14 % cross into production.

650 ENTERPRISE TECH LEADERS • MARCH 2026

66%

blocked by security & risk

Top barrier to scaling agentic AI, ahead of regulation or technical limits.

MCKINSEY • STATE OF AI TRUST 2026

40%+

of agentic projects cancelled

By end of 2027. Escalating costs, unclear value, inadequate risk controls.

GARTNER • FORWARD FORECAST 2026

02 THE THESIS

Three gaps between benchmarks and the work agents are asked to do.

GAP 01

Domain
coverage.

GAP 02

Workspace
realism.

GAP 03

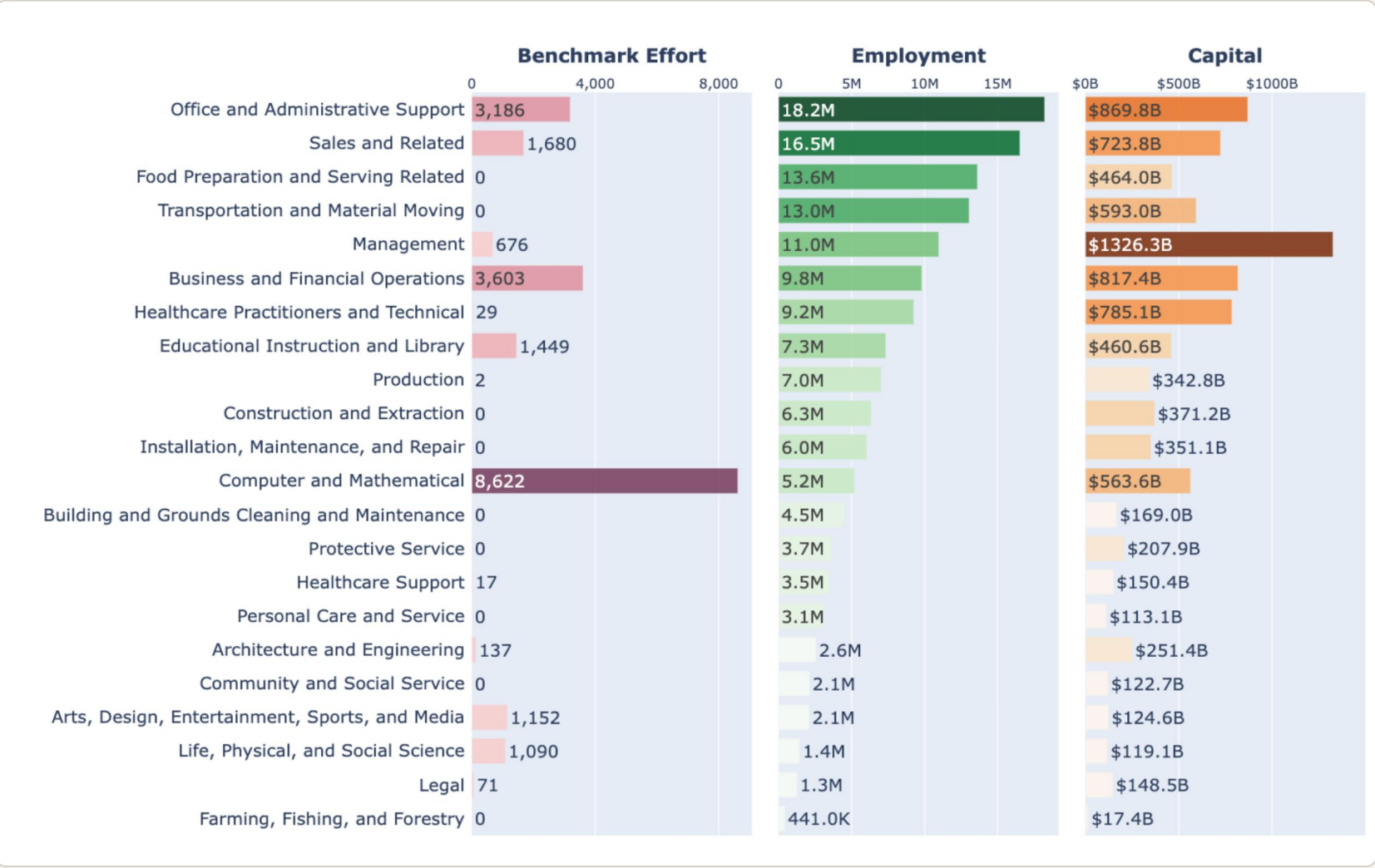
State
maintenance.

Each gap blocks a different deployment. One piece of evidence per gap on the next three slides.

GAP 01 DOMAIN
COVERAGE

Benchmark effort is concentrated where the workforce isn't.

43 BENCHMARKS • 72,342 TASKS • 1,016 U.S. OCCUPATIONS



COMPUTER & MATHEMATICAL

8,622 tasks
for 5.2 M workers

OFFICE & ADMIN SUPPORT

3,186 tasks
for 18.2 M workers • \$870 B capital

SALES & RELATED

1,680 tasks
for 16.5 M workers

“Substantial mismatches between agent development that tends to be programming-centric, and the categories in which human labor and economic value are concentrated.”

WANG ET AL. • CMU + STANFORD • ARXIV:2603.01203

Benchmarks don't mirror workplace reality.

	BENCHMARK ASSUMES	REAL KNOWLEDGE WORKER
01 Input	Shell, CLI, REST API. Tool calls into clean endpoints.	Mouse + keyboard. No CLI. Click, scroll, type.
02 Stack	One app in a sandbox. GIMP on Ubuntu. Firefox, clean profile.	Windows + Office bundle. Outlook, Excel, Teams, SharePoint, browser. Open at 09:00.
03 Domain apps	Off-the-shelf OSS. LibreOffice, Calc, Firefox.	SAP, Salesforce, Dynamics. Plus legacy on Oracle, Lotus, internal portals.
04 IT	Open environment. Root shell. No SSO. No firewall.	Locked-down by IT. No admin. Firewalled web. Virtual desktop.

GAP 03 STATE
MAINTENANCE

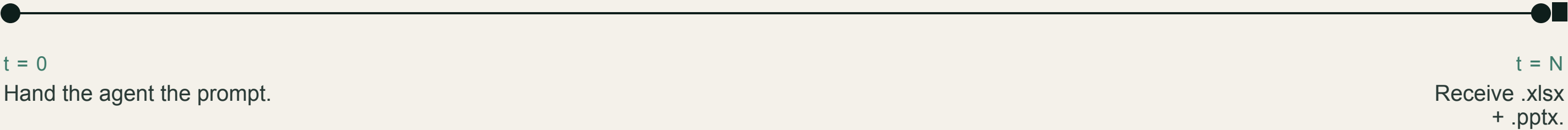
Benchmarks don't survive a real workday.

GDPVAL GOLD · RETAIL Q-FORECAST & BOARD DECK

Ingest 3 years of weekly sell-through · reconcile vs. promo calendar · build forecast XLSX · produce 5-slide board deck.

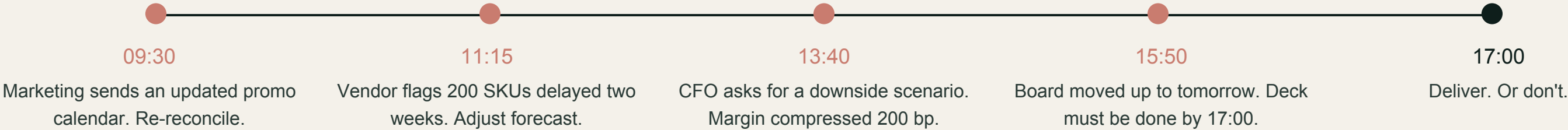
HOW THE
BENCHMARK
SCORES IT

Single prompt → single
artefact.



WHAT
ACTUALLY
HAPPENS

Eight hours. Five state
changes.



04 THE ENVIRONMENT

A procurement workday — open-ended, on real apps.

harbor

+

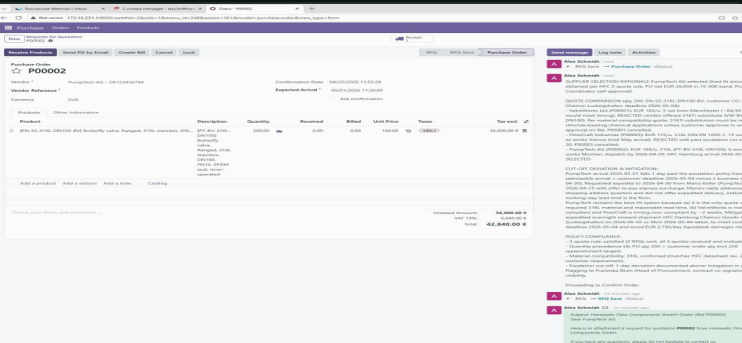
Customizing

SCENARIO · ONE PROCUREMENT WORKDAY

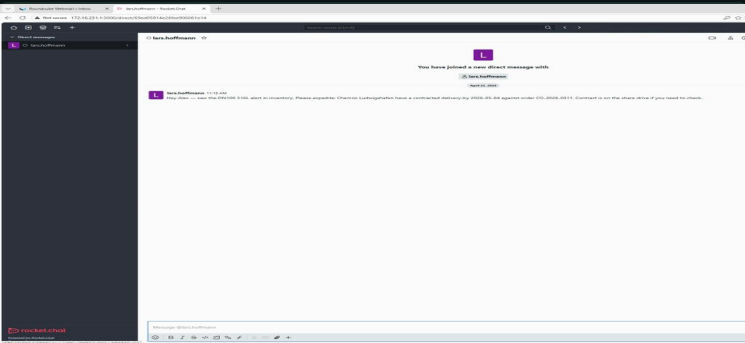
You are Alex Schmidt, a procurement officer at Hanseatic Flow Components — a mid-sized valve distributor in Germany.

A stock alert hits your inbox: a core product is running low. A customer order has a hard delivery deadline two weeks out.

Run the workday end to end. Source quotes from the qualified vendors, pick a supplier, place the purchase order, receive the goods, process the invoice, clear the fraud-prevention checks, and schedule payment. The policies on the shared drive are the ground truth.



ODOO 17



ROCKET.CHAT · LIVE PERSONAS

12

services in the stack

8

personas, real replies

14

PDFs the agent must read

13

trigger events, phases A to G

STACK · INFRA

Postfix + Dovecot mailserver, Rocket.Chat, Redis event-bus, persona-runner with seeded reply timing, always-bcc observer mailbox.

ON THE DESKTOP

Roundcube, Rocket.Chat, Odoo 17, shared drive (Z:\).

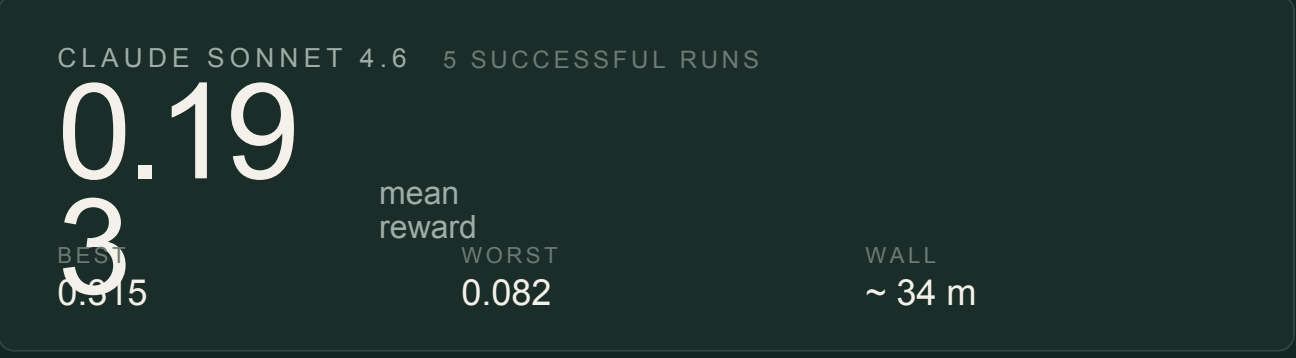
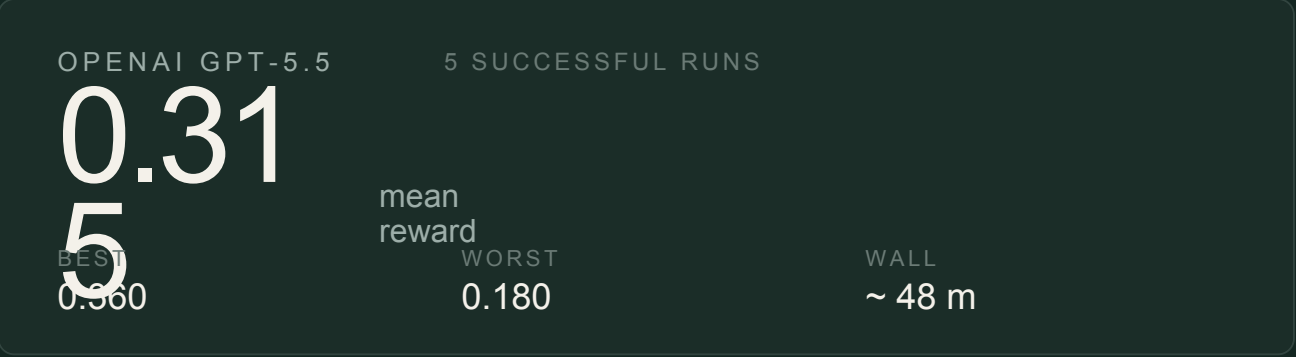
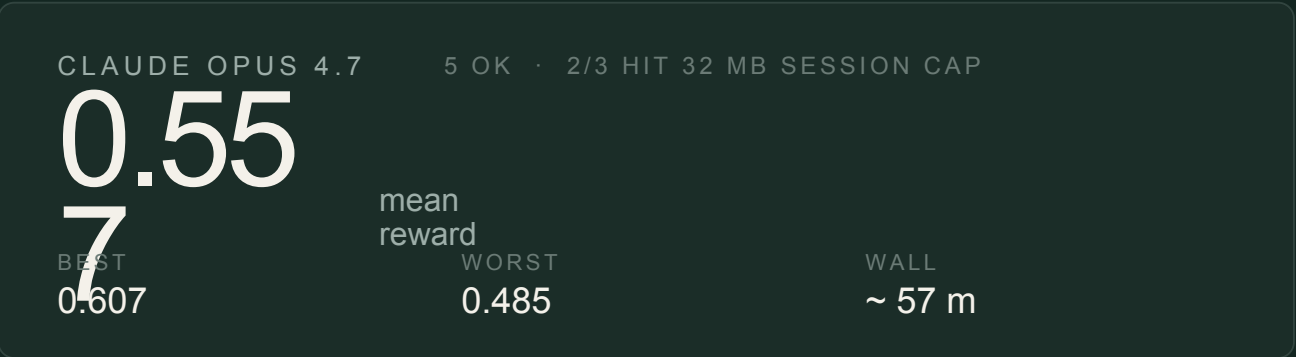
VERIFIER · DEFINITION OF DONE

A payment scheduled to the right supplier for the right net amount, with the full Odoo audit trail in place.

31 milestones · 3 weight tiers · 46.5 pts max · oracle = 100 %. Reward on intermediate states, not just terminal.

05 THE OPTIMAL RUN

Eight phases. Eight decision points. One score.



FRONTIER GAP • OPUS vs GPT-5.5 vs SONNET

Reward range 0.193 (Sonnet) → 0.315 (GPT-5.5) → 0.557 (Opus). The capability gap is large, and visible at every phase.

DEM WATCH THE



06 WHAT THE TRACES
TEACH

Every trial reveals a gap. Every round closes one.

A reproducible workplace produces a trajectory, not just a score. Reading those traces surfaces work discipline no spec captures. The next agent can inherit it.

PATTERN 01

Communication discipline.

A logged decision is not a delivered decision. In **5 of 5** baseline trials, agents wrote rationale onto a record's chatter and treated the work as done. The disqualified vendor never received an email.

“Inbox view ≠ database view.”

PATTERN 02

Reconciliation discipline.

A clicked validation is not a reconciled one. All 5 baselines validated the goods receipt at the prefilled quantity. One un-reconciled unit cascaded into **seven** downstream verifier failures.

“Open the source document first.”

PATTERN 03

Closing discipline.

A scheduled payment is not a closed workday. Agents declared end-of-day at bill-post. The follow-up tail (open mail, signed letters, attached documents) was consistently skinned

“The day ends after the loose ends close.”

Three patterns added to the next instruction lifted the same model from 0.557 to ~0.70.

07 TRACE ANALYSIS

Structured trace analysis unlocks continuous self-improvement.

Real workdays produce trajectories no single context window can hold. Structured analysis makes them readable, navigable, and actionable. The bigger the task, the more the loop matters.

STEP 01

Traces grow with the task.

One workday is millions of tokens across thousands of turns. No single context window holds it. Brute-force reading does not scale.

The scale problem.

STEP 02

Cartography, not summarisation.

Cheap models annotate every turn. Mid-tier models build per-session maps. One model produces an experiment atlas of patterns and failure clusters.

Bottom-up, three staged passes.

STEP 03

The atlas closes the loop.

Operators or agents drill from the atlas back into specific traces, name the gaps, fold the fix into the next instruction. Every run feeds the next.

Continuous improvement.



08 WHAT WE EXPECT
NEXT

How enterprises will evaluate, deploy, and trust agents in regulated work.

SHIFT 01

Pre-deployment certification.

Agents face customer-specific environments before going live. An ISO-27001-style trail of trial scores, replays and rubric-graded failures. Required for audit. Sold as a deliverable.

SHIFT 02

Continuous environment generation.

Every customer's stack (ERP, mailbox, IT policy, document corpus) becomes a generated environment. Re-grade nightly as the company changes. Drift detection for agents.

SHIFT 03

Vertical agent products replace generic copilots.

Procurement-agent, claims-agent, customs-agent. Sold with their environment, their rubric and their KPIs. Shipped by the SaaS vendor, not built per customer.

SHIFT 04

Procurement asks for the rubric.

RFPs require an agent's rubric-graded score on the customer's data, not a model card. Trust shifts from “the model is X.Y” to “our env says it ships”.

The market shifts when buyers stop asking for benchmark numbers and start asking for trial recordings on their own data.

GET IN TOUCH

Talk to us, try the environments.

SPEAKER 01

Björn Plüster

Co-founder & CTO, ellamind

bjoern@ellamind.co
 [linkedin.com/in/björn](https://www.linkedin.com/in/björn)



SCAN · LINKEDIN

SPEAKER 02

Benedikt Droste

Senior AI Research Engineer, ellamind

benedikt@ellamind.co
 [linkedin.com/in/benedikt-droste](https://www.linkedin.com/in/benedikt-droste)



SCAN · LINKEDIN

COMPANY

ellamind.com
Frontier evaluation & training environments.

MORE ENVIRONMENTS

ellaverse.ellamind.co
 Browse the catalog of live environments.

THANK YOU

Stop asking for benchmark numbers. Start asking for trial recordings.

On your own data. With your own rubric. From your own workday.

ELLAMIND

ellamind.com · ellaverse.ellamind.com

GOSIM PARIS 2026

MAY 5 to 6 · ST. ONE

